

Four-Element MacNeille G2 and Finite APS Witness

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Four-Element MacNeille Witness

Pass 113 extends the MacNeille G2/A2 boundary search from the fixed three-element V-carrier to all labelled four-element posets with a unique bottom. The result is positive: the Pass-112 A2 gate is a three-element artifact.

The explicit witness has carrier $\{0, a, b, c\}$ with order

$$0 < a < b, \quad 0 < c,$$

and no comparison between c and a, b beyond $0 < c$. Set $\tau = a$, $\text{bottom} = 0$,

$$\boxtimes(0) = \boxtimes(a) = b, \quad \boxtimes(b) = \boxtimes(c) = 0,$$

and

$$\square(0) = \square(a) = \square(c) = 0, \quad \square(b) = b.$$

This table has no syntactic `boxtimes` fixed point. Under the v1 MacNeille extension, the whole cut $\{0, a, b, c\}$ is fixed and non-principal. The standalone checker report confirms finite A1-A4, G2, and FG2:

- A2 holds because $a \leq \text{boxtimes}(0) = b$.
- G2 holds because $\text{boxtimes}(a) = b$ is not below 0 .
- FG2 holds because $\text{boxtimes}(\text{boxtimes}(a)) = 0 \leq b = \text{boxtimes}(a)$.

The labelled-poset enumeration found:

package	table count	profiles	posets
completion separation	359424	1404	40
separation + G2	135168	528	40
separation + G2 + A2	61440	240	36
separation + G2 + A124Core	3024	240	36
separation + G2 + A1-A4 APS	2784	240	36

Thus G2, FG2, and finite A1-A4 table checks alone do not force syntactic reflection of MacNeille completion fixed cuts. The next boundary is residual and completion stability: can this witness carry a compatible tensor/residual package, and do the relevant APS laws survive completion?

Machine artifacts: `code/scripts/check-pass113.py`, `code/models/examples/four-element-g2-aps-nosynt.json`, `artifacts/reports/pass113-four-element-macneille-g2-boundary-check.json`, and `artifacts/reports/macneille-reflection-four-element-g2-aps-nosynt-v1.json`.